

URVASHI JAGDISH GAWAS

urvashijgawas@gmail.com linkedin.com/in/urvashi-gawas/ [+91 9769003513](tel:+919769003513) [Github](#)

EDUCATION

Savitribai Phule Pune University

M.Sc Bioinformatics, Bioinformatics Centre, Savitribai Phule Pune University, Pune, India

08/2024 – 05/2026

CGPA: 7.94/10

Mumbai University

B.Sc Zoology, D.G. Ruparel College, Mumbai, India

08/2020 – 05/2023

CGPA: 8.60/10

EXPERIENCE

Evolvus Inc., Bioinformatics Intern, Pune, India

06/2025 – 07/2025

- Segmented immune regions in H&E histology images using QuPath to delineate tissue architecture.
- Analyzed 10x Visium spatial transcriptomics data and used SEAGAL (L-index) to quantify spatial co-localization of B cells, T cells, and macrophages.
- Integrated Squidpy and GSEAPy to identify spatial gene modules and B cell-associated inflammatory signatures.

Charvi Care Clinic, Social Media Intern, Mumbai, India

11/2021 – 02/2022

- Managed content creation and scheduling for a dermatology and hair transplantation clinic, maintaining consistent brand presence across platforms.
- Monitored audience engagement metrics to inform post timing and content strategy.

RESEARCH PROJECTS

Computational Drug Discovery in Autosomal Dominant Polycystic Kidney Disease

2025

- Performed molecular docking of FDA-approved drugs against the P2RX7 receptor using Schrödinger Maestro to identify high-affinity binders.
- Conducted molecular dynamics simulations (GROMACS) to evaluate stability of protein–ligand complexes under physiological conditions.
- Performed similarity-based virtual screening targeting the mTOR signalling pathway to identify potential inhibitors.

Machine Learning Pipeline for Ovarian Cancer Subtype Classification

2025

- Built a computational framework to classify molecularly distinct ovarian cancer subtypes from RNA-seq transcriptomic data using PCA for dimensionality reduction and Ant Colony Optimization for gene feature selection.
- Benchmarked SVM, Logistic Regression, Random Forest, and XGBoost classifiers on a 90:10 train-test split; designed workflow for future integration into a GUI-based subtype prediction tool.

SKILLS

Programming: Python, R, SQL, HTML, CSS, Linux/Bash

Machine Learning: scikit-learn, PCA, feature selection, classification pipelines

Genomics & Transcriptomics: BWA, Bowtie2, SAMtools, GATK, STAR, DESeq2, Nextflow, Trimmomatic, SEAGAL

Spatial & Image Analysis: QuPath (tissue segmentation), SEAGAL (spatial analysis)

Molecular Modelling: Schrödinger Maestro, GROMACS, AutoDock, PyMOL, PyRx

ACHIEVEMENTS

GATB 2024 – DBT Funded Studentship

2024

- Qualified the Graduate Aptitude Test in Biotechnology (GATB); selected for a 2-year DBT-funded research studentship.

CONFERENCES & CERTIFICATIONS

National Conference on Emerging Trends in Bioinformatics & Computational Biology

2026

- Presented poster “*In Silico Docking Study of AVPR2 & mTOR as Therapeutic Targets in ADPKD*”; demonstrated molecular docking and virtual screening approaches.

PacBio Bioinformatics Seminar

- Attended seminar on long-read sequencing data analysis and bioinformatics applications.

Illumina DRAGEN Solutions for Viral Genome Analysis

- Completed training on accelerated secondary analysis pipelines for viral genomics using Illumina DRAGEN.

Python Programming – Techgen Computer Training Institute

- Completed foundational Python course covering data structures, scripting, and automation.